

State observers are unnecessary for induction motor control

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Abstract Induction motors constitute a theoretically interesting and practically important class of nonlinear systems, which are evolving into a benchmark example for nonlinear control. They are described by a fifth-order nonlinear differential equation with two inputs, and only three state variables available for measurement. A lot of research in the field has been devoted to the design of observers which, combined with a suitable control strategy, would yield stable behaviour. Our contribution in this paper is to show that the control objective can be achieved without reconstruction of the motor state. Specifically, we present a globally stable nonlinear dynamic output feedback controller for torque tracking of induction motors which does not rely on state reconstruction ideas. Another important feature of our scheme is that the control law is globally defined, even in startup. This stems from the fact that we do not aim at linearizing the system dynamics, but instead exploit the energy dissipation properties of the motor model. For the sake of illustration we present the result for a model described in the stator frame (*ab* model), but the theory applies as well to models expressed in a rotating frame (*dq* model). We also show how, as a particular case of torque tracking, we can solve the rotor speed tracking problem.

Keywords Induction motor, observers, nonlinear systems, stabilization

1. Problem formulation

We consider in this paper the well-known *ab* model of a three-phase induction motor [14],

$$D\dot{x} + C(x)x + Rx = Q, \quad (1.1)$$

where the state vector is given by the *a* and *b* components of the stator currents (i_{sa}, i_{sb}) and the rotor fluxes (ψ_{ra}, ψ_{rb}) respectively, and the rotor speed (ω), that is

$$x \triangleq [i_{sa}, i_{sb}, \psi_{ra}, \psi_{rb}, \omega]^T \triangleq [x_1, x_2, x_3, x_4, x_5]^T,$$

with

$$D \triangleq \begin{bmatrix} L_r \sigma \mathcal{J}_2 & 0 & 0 \\ 0 & \mathcal{J}_2 & 0 \\ 0 & 0 & L_r J \end{bmatrix}, \quad C(x) \triangleq \begin{bmatrix} 0 & 0 & f_1(x) \\ 0 & -x_5 \mathcal{J}_2 & 0 \\ -f_1^T(x) & 0 & 0 \end{bmatrix},$$

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$$R \triangleq \begin{bmatrix} L_r \sigma_r \mathcal{I}_2 & -\frac{MR_r}{L_r} \mathcal{I}_2 & 0 \\ -\frac{MR_r}{L_r} \mathcal{I}_2 & \frac{R_r}{L_r} \mathcal{I}_2 & 0 \\ 0 & 0 & L_r B \end{bmatrix}, \quad Q \triangleq \begin{bmatrix} L_r u_1 \\ L_r u_2 \\ 0 \\ 0 \\ 0 \\ -L_r T_L \end{bmatrix},$$

$$f_1(x) \triangleq \begin{bmatrix} -Mx_4 \\ Mx_3 \end{bmatrix} = M \mathcal{I}_2 \begin{bmatrix} x_3 \\ x_4 \end{bmatrix}, \quad \mathcal{I}_2 \triangleq \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \sigma \triangleq L_s - \frac{M^2}{L_r}, \quad \gamma \triangleq \frac{R_s L_r^2 + R_r M^2}{\sigma L_r^2},$$

where $\mathcal{I}_2$ is the 2×2 identity matrix, R_s, R_r, L_s, L_r are the stator and rotor resistances and inductances, respectively, M is the mutual inductance, u_1, u_2 are the stator voltages, and T_L is the load torque

The control problems we solve in this paper may be formulated as follows

Torque tracking problem. Consider the induction motor model (1.1) with control inputs the stator voltages u_1, u_2 and regulated output the generated torque

$$T \triangleq h(x) \triangleq \frac{M}{L_r} (x_2 x_3 - x_1 x_4). \quad (1.2)$$

Assume:

- (A1) Stator currents x_1, x_2 and rotor speed x_5 are available for measurement.
- (A2) Motor parameters are exactly known
- (A3) The desired torque T_d is a smooth bounded differentiable function with a known bounded first-order derivative and the load torque T_L is smooth and bounded.

Under these conditions, design a control law that will ensure

$$\lim_{t \rightarrow \infty} (T - T_d) = 0 \quad (1.3)$$

with all internal signals bounded

Rotor speed tracking problem. For the problem formulation above assume (A1), (A2) and the following

- (A3') The desired rotor speed x_{5d} is a smooth bounded differentiable function with known bounded first- and second-order derivatives
- (A4) Load torque can be linearly parametrized as

$$T_L = p^T \phi \quad (1.4)$$

where $p \in \mathbb{R}^q$ is an unknown constant vector with $\phi, \dot{\phi}$ bounded measurable signals. Further, we assume known an upperbound K_p on the norm of the parameter vector, i.e., $\|p\| \leq K_p$, where $\|\cdot\|$ stands for the Euclidean norm

Under these conditions, design a control law that will ensure

$$\lim_{t \rightarrow \infty} (x_5 - x_{5d}) = 0 \quad (1.5)$$

with all internal signals bounded

Remark 1.1. The model (1.1) coincides with [8, eq. (16)], with the exception of the mechanical damping constant $B > 0$, which is set to zero in [8]. As will become clear later, the existence of nonzero mechanical damping will be essential to solve the rotor speed tracking problem. The need for this assumption stems from the inability to add damping to the mechanical system with partial state measurement. See [11] for further discussion on this point, and [10] for physical motivation of the chosen motor representation.

Remark 1.2. It is worth pointing out that from the point of view of modern applications of induction motors (e.g. electric vehicles, robotics), the regulated output of interest is torque. We refer the reader to [10, 11] for comments regarding the assumptions and further motivation for these problems

Remark 1.3. It has been shown in [10] that the induction motor model with zero load torque T_L defines a passive mapping [3] from stator voltages u_1, u_2 to stator currents. This property follows immediately taking the time derivative of the motor total energy $x^T D x$ along the trajectories of (11), and using the positivity of D , R and the skew-symmetry of $C(x)$. We will find it convenient below to decompose this passive system into the feedback interconnection of two passive subsystems. This decomposition reveals the interaction of the subsystems and will allow us to solve our stabilization problem

2. Literature review and paper contributions

A brief review of the literature follows. The problem of torque regulation assuming full state measurement was studied using linearization techniques by Deluca [2] for a model neglecting the mechanical dynamics, i.e. $x_5 = \text{const}$. Marino et al. [8] propose an adaptive version of the feedback linearization scheme of [7] to address the speed regulation problem with unknown rotor resistance R_r and unknown constant load torque, but assuming measurable state. Kanellakopoulos et al. [6] established local stability of a scheme designed using backstepping, which is a recent Lyapunov-based stabilization technique, for the velocity control problem with flux observer and known motor parameters.

All the schemes referred above suffer from the drawback that the control laws are not globally defined. The set of singularities, besides being difficult to analyse, is defined by the design methodology. The physical rationalization of these singularities is usually an a posteriori step, leaving the designer with a pious hope that it will not contradict physical operation. This seems to stem from the fact that both techniques, feedback linearization and backstepping, neglect the physical structure of the system. In contrast with these approaches, an energy shaping plus damping injection controller design methodology that exploits the energy dissipation properties of the system has been proposed in [9]¹. This methodology is used in [10] to derive the first globally defined globally stable solution to the torque regulation problem with partial state feedback and unknown constant load. The result was established for constant desired torques which satisfy an upper bound determined by the motor mechanical damping. An adaptive scheme to handle uncertain rotor resistance is also presented in that paper, but requires full measurement of the state. The result of [10] is extended in [11] to handle time-varying torque references, and to remove the upper-bound condition described above. The key idea introduced in [11] is a procedure to inject damping to the mechanical subsystem. In [1] a local solution to the problem of position control of robots with induction motor drives is presented.

The schemes of [10, 11] are derived on the basis of a model where the variables are expressed in a synchronously rotating frame (dq model). In [5] it is shown that, for the case of full state measurement, we can rotate back the coordinates of the control signals of the scheme of [11] to solve the torque tracking problem for a motor model given in the fixed stator frame (ab model). This ‘coordinate-invariance’ feature of the controller of [11] is not surprising since the design procedure of [9] is based on the input–output (energy dissipation) properties of the motor, which are ‘coordinate-independent’.

It is not known at this point if the coordinate rotation used in [5] applies also for the observer-based scheme of [11]. In any case, it is clear that it will lead to a highly complex controller dynamics. One of the contributions of the present work is to provide a very simple ‘observerless’ solution to the torque tracking problem above. To this end, we make the fundamental observation that the mechanical part of the induction motor dynamics defines a passive feedback around the electrical subsystem, which in turn is also passive. Therefore, we can apply the energy shaping procedure only to the electrical part and treat the effect of the

¹ The methodology is a natural extension of the highly successful passivity-based controller designs used in robotics, see e.g. [12]

mechanics as a passive perturbation. Interestingly enough this can be done measuring only stator variables. The main stumbling block of this procedure is that we cannot inject additional damping to the mechanical subsystem. This dependence on the system's natural damping limits the achievable performance of the design.

3. Motor model revisited

In this section we explain the rationale of the controller that we propose. To this end, we find it convenient to express the induction motor model as a feedback interconnection of passive systems as follows. Model (1.1) can be rewritten in an alternative form as

$$D_e \dot{x}_e + C_e(x_5)x_e + R_e(x_5)x_e = Q_e, \quad (3.1)$$

$$J \dot{x}_5 + Bx_5 = T - T_L, \quad (3.2)$$

where

$$D_e \triangleq \begin{bmatrix} L_r \sigma \gamma \mathcal{J}_2 & 0 \\ 0 & \mathcal{J}_2 \end{bmatrix}, \quad C_e(x_5) \triangleq \begin{bmatrix} 0 & M \mathcal{J}_2 \\ M \mathcal{J}_2 & -\mathcal{J}_2 \end{bmatrix} x_5, \\ R_e(x_5) \triangleq \begin{bmatrix} L_r \sigma \gamma \mathcal{J}_2 & -\frac{MR_r}{L_r} \mathcal{J}_2 \\ -M \mathcal{J}_2 x_5 - \frac{MR_r}{L_r} \mathcal{J}_2 & \frac{R_r}{L_r} \mathcal{J}_2 \end{bmatrix}, \quad x_e \triangleq \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \quad Q_e \triangleq \begin{bmatrix} L_r u_1 \\ L_r u_2 \\ 0 \\ 0 \end{bmatrix}.$$

Notice that to obtain a skew-symmetric² $C_e(x_5)$ we have added and subtracted $M \mathcal{J}_2 x_5$ in the third and fourth rows of (3.1). This has in its turn added a term $-M \mathcal{J}_2 x_5$ in the (2,1)-entry of the 'damping matrix' $R_e(x_5)$. As shown below, we will be able to recover the positivity of this matrix using control signals that compensate this term. (3.1), (3.2) define two operators in a feedback interconnection

$$\Sigma_1: v_1 \triangleq \begin{bmatrix} u_1 \\ u_2 \\ -x_5 \end{bmatrix} \rightarrow y_1 \triangleq \begin{bmatrix} x_1 \\ x_2 \\ T \end{bmatrix}, \\ \Sigma_2: (-T + T_L) \rightarrow -x_5 = \frac{1}{J_s + B} (-T + T_L), \quad s \triangleq \frac{d}{dt},$$

where T_L is viewed as an external disturbance. Notice that

$$c_1 \triangleq \lambda_{\min} \left\{ \begin{bmatrix} L_r \sigma \gamma \mathcal{J}_2 & -\frac{MR_r}{L_r} \mathcal{J}_2 \\ -\frac{MR_r}{L_r} \mathcal{J}_2 & \frac{R_r}{L_r} \mathcal{J}_2 \end{bmatrix} \right\} > 0$$

with $\lambda_{\min}\{\cdot\}$ the minimum eigenvalue. Now, taking the derivative of the total energy of the electrical subsystem $x_e^T D_e x_e$ along the trajectories of (3.1), we can show that Σ_1 is passive [3]. Specifically, it satisfies the dissipation inequality

$$\int_0^t v_1^T(\tau) y_1(\tau) d\tau \geq c_1 \int_0^t \|x_e(\tau)\|^2 d\tau + c_2$$

² As discussed in [12] a representation that enjoys this property reveals the workless forces in the system. Since these forces do not affect the system stability there is no need to cancel them.

for all $t \geq 0$ and some $c_2 \in \mathbb{R}$. On the other hand, the system Σ_2 is strictly positive real for all $J \geq 0, B > 0$; thus it is also passive. Motivated by the well-known fact that feedback interconnection of passive subsystems is also passive, we then propose to design an energy shaping plus damping injection controller for the electrical subsystem only, and to treat the mechanical subsystem as a ‘passive disturbance’. This is in stark contrast to the approach of [10], where the controller aims at shaping the energy of the overall system. Interestingly enough, the energy shaping of the electrical subsystem and the damping injection can be carried out using only measurable variables as shown below

4. Main results

Proposition 4.1 (Torque tracking). *Consider the induction motor model (1.1), (1.2) under assumptions (A1)–(A3). Let the controller be defined as*

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \sigma \begin{bmatrix} \dot{x}_{1d} \\ \dot{x}_{2d} \end{bmatrix} + \sigma \gamma \begin{bmatrix} x_{1d} \\ x_{2d} \end{bmatrix} + \left(\frac{M}{L_r} x_5 \mathcal{J}_2 - \frac{MR_r}{L_r^2} \right) \begin{bmatrix} x_{3d} \\ x_{4d} \end{bmatrix} - L_r K(x_5) \begin{bmatrix} (x_1 - x_{1d}) \\ (x_2 - x_{2d}) \end{bmatrix}, \quad (4.1)$$

where

$$x_{ed} \triangleq \begin{bmatrix} x_{1d} \\ x_{2d} \\ x_{3d} \\ x_{4d} \end{bmatrix} = \begin{bmatrix} \beta \mathcal{J}_2 + \frac{L_r}{\beta M^2} T_d \mathcal{J}_2 \\ M \beta \mathcal{J}_2 \end{bmatrix} z, \quad \beta > 0, \quad (4.2)$$

with $z = [z_1, z_2]^T$ the two-dimensional controller state solution of

$$\dot{z} = \left(\frac{R_r}{\beta^2 M^2} T_d + x_5 \right) \mathcal{J}_2 z, \quad (4.3)$$

for some initial condition³ $z(0) = [\cos a, \sin a]^T$, $a \in \mathbb{R}$, and $K(x_5)$ given as

$$K(x_5) = \frac{M^2}{4\varepsilon} x_5^2, \quad 0 < \varepsilon < R_r. \quad (4.4)$$

Under these conditions,

$$\lim_{t \rightarrow \infty} (T - T_d) = 0$$

with all internal signals bounded.

Proposition 4.2 (Speed tracking). *Consider the induction motor model (1.1), (1.2) under assumptions (A1), (A2), (A3'), (A4). Let the controller be defined by (4.1)–(4.4), with desired torque*

$$T_d = \hat{p}^T \phi + J \dot{x}_{5d} + B x_{5d} \quad (4.5)$$

$\hat{p}$ is updated with a gradient update law with projection

$$\dot{\hat{p}} = \text{Proj} \{ \hat{p} - g \phi(x_5 - x_{5d}) \}, \quad \hat{p}(0) = p_0 \in \mathbb{R}^q, \quad g > 0 \quad (4.6)$$

where $\text{Proj} \{ \cdot \}$ is a smooth projection⁴ that keeps the estimates inside a sphere of radius K_p .

Under these conditions,

$$\lim_{t \rightarrow \infty} (x_5 - x_{5d}) = 0$$

with all internal signals bounded

³ The motivation for this particular choice of initial conditions will become clear in the proof of the proposition

⁴ See e.g. [13] for an example of such a projection

Remark 4.1. Notice that the controllers of the propositions above are nonlinear dynamic feedbacks of the measurable variables x_1, x_2, x_5 , where the controller states are z and $\hat{p}$. As will become clear in the proofs, there are two types of terms in (4.1): the first three right-hand terms achieve the energy shaping of the closed loop, while the last right-hand terms inject the required damping to achieve asymptotic stability.

Remark 4.2. The control (4.1) requires the knowledge of $\dot{x}_{1d}, x_{2d}$, which in turn implies availability of $\dot{T}_d$. This hampers the inclusion into the definition of T_d of the actual rotor speed x_5 without acceleration measurement. Also, this explains the need for our assumption of known $\dot{\phi}$ in the speed tracking problem formulation.

5. Proof of main results

5.1 Proof of Proposition 4.1

The proof proceeds along the following steps. First, we define a change of coordinates (an error signal) for the electrical variables x_e such that convergence to zero in these coordinates (with bounded states) implies the torque tracking objective. Then, we derive the dynamic equations in the new coordinates and prove that, if there is no finite escape time, then convergence to zero is achieved. Finally, we prove the existence of solutions for the whole positive real axis⁵.

Step 1. Let us define an error signal $\tilde{x}_e \triangleq [\tilde{x}_1, \tilde{x}_2, \tilde{x}_3, \tilde{x}_4]^T \triangleq x_e - x_{ed}$, where x_{ed} is given by (4.2). Now, by direct substitution we can prove that the solutions to (4.3) may be written in the form

$$z_1 = \cos \theta, \quad z_2 = \sin \theta,$$

where θ is the solution of

$$\dot{\theta} = \frac{R_r}{\beta^2 M^2} T_d + x_5, \quad \theta(0) = a$$

Using the expression above, replacing (4.2) in (1.2), and after some lengthy but straightforward calculations we establish the key property of

$$h(x_{ed}) \equiv T_d$$

On the other hand, it is easy to show that

$$T - T_d = h(x) - h(x_{ed}) = \frac{M}{2L_r} (\tilde{x}_e^T W \tilde{x}_e + 2\tilde{x}_e^T W x_{ed})$$

with

$$W \triangleq \begin{bmatrix} 0 & \mathcal{J}_2 \\ -\mathcal{J}_2 & 0 \end{bmatrix}$$

Therefore, noting that W is a unitary matrix, the bound below follows immediately⁶

$$|T - T_d| \leq \frac{M}{2L_r} (\|\tilde{x}_e\|^2 + 2\|\tilde{x}_e\| \|x_{ed}\|). \quad (5.1)$$

From (5.1) we conclude that asymptotic torque tracking will be achieved if we can ensure that $\tilde{x}_e \rightarrow 0$ as $t \rightarrow \infty$ with bounded x_{ed} .

⁵ It is worth mentioning that the only technically involved part of the proof, which imposes some of the restrictions in our problem formulation, is this last step.

⁶ Notice that $h(\cdot)$ is not globally Lipschitz.

Step 2· Writing (3.1) in terms of $\tilde{x}_e$ we get

$$D_e \dot{\tilde{x}}_e + C_e(x_5) \tilde{x}_e + R_e(x_5) \tilde{x}_e = \xi,$$

where

$$\xi \triangleq Q_e - [D_e \dot{x}_d + C_e(x_5) x_d + R_e(x_5) x_d]$$

which can be further expressed as

$$\xi = \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \\ \xi_4 \end{bmatrix} = \begin{bmatrix} L_r u_1 - L_r \sigma \dot{x}_{1d} + M x_5 x_{4d} - L_r \sigma \gamma x_{1d} + \frac{M R_r}{L_r} x_{3d} \\ L_r u_2 - L_r \sigma \dot{x}_{2d} - M x_5 x_{3d} - L_r \sigma \gamma x_{2d} + \frac{M R_r}{L_r} x_{4d} \\ - \dot{x}_{3d} - x_5 x_{4d} + \frac{M R_r}{L_r} x_{1d} - \frac{R_r}{L_r} x_{3d} \\ - \dot{x}_{4d} + x_5 x_{3d} + \frac{M R_r}{L_r} x_{2d} - \frac{R_r}{L_r} x_{4d} \end{bmatrix}$$

Direct replacement of (4.1) shows that

$$\begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = -K(x_5) \mathcal{J}_2 \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix}. \quad (5.2)$$

Now, from (4.2) and (4.3) we have

$$\begin{aligned} \begin{bmatrix} \dot{x}_{3d} \\ \dot{x}_{4d} \end{bmatrix} &= \left(\frac{R_r}{\beta^2 M^2} T_d + x_5 \right) \mathcal{J}_2 \begin{bmatrix} x_{3d} \\ x_{4d} \end{bmatrix}, \\ \begin{bmatrix} x_{1d} \\ x_{2d} \end{bmatrix} &= \frac{1}{M} \left(\mathcal{J}_2 + \frac{L_r}{\beta^2 M^2} T_d \mathcal{J}_2 \right) \begin{bmatrix} x_{3d} \\ x_{4d} \end{bmatrix}. \end{aligned} \quad (5.3)$$

Thus, after some lengthy but simple calculations we can prove that $\xi_3 = \xi_4 = 0$. This leads finally to the error equation

$$D_e \dot{\tilde{x}}_e + C_e(x_5) \tilde{x}_e + R_{es}(x_5) \tilde{x}_e = 0, \quad (5.4)$$

where

$$R_{es}(x_5) \triangleq \begin{bmatrix} L_r [\sigma \gamma + K(x_5)] \mathcal{J}_2 & -\frac{1}{2} M \mathcal{J}_2^T x_5 - \frac{M R_r}{L_r} \mathcal{J}_2 \\ -\frac{1}{2} M \mathcal{J}_2 x_5 - \frac{M R_r}{L_r} \mathcal{J}_2 & \frac{R_r}{L_r} \mathcal{J}_2 \end{bmatrix}.$$

In view of (5.4) we say that the total energy of the subsystem Σ_1 in a closed loop with the control law has been shaped to match the function

$$H_d \triangleq \frac{1}{2} \tilde{x}_e^T D_e \tilde{x}_e$$

with external forces $R_{es}(x_5) \tilde{x}_e$. The terms (5.2) inject the required damping into the error system, since it is easy to verify that with the choice of $K(x_5)$ given in (4.4) we have

$$\lambda_{\min} \{R_{es}(x_5)\} \geq \alpha > 0$$

Now, let us temporarily assume that x_5 cannot escape to infinity in finite time. Under this condition, we can evaluate the derivative of H_d along the trajectories of (5.4) to get

$$\dot{H}_d = -\tilde{x}_e^T R_{es}(x_5) \tilde{x}_e.$$

Thus, we can determine positive constants m_e, ρ_e , which are independent of t , such that

$$\|\tilde{x}_e(t)\| \leq m_e \|\tilde{x}_e(0)\| e^{-\rho_e t}, \quad \forall t \quad (5.5)$$

The only remaining difficulty is thus to prove that there is no finite escape time behaviour.

Step 3 To this end, notice that the dynamics of the closed-loop system are fully described by (5.4), (3.2) and (1.2). Notice that this set of differential equations is locally Lipschitz in the state and, under the assumptions on the reference, is also continuous in t . Therefore, there exists a time interval $[0, t_1]$ where the solutions exist and are unique. Now, for all $t \in [0, t_1]$, (5.5) holds. This, together with the boundedness of $\|x_{ed}\|$ (4.2), proves that on the time interval $[0, t_1]$, the right-hand side of (3.2) is bounded by a constant α_1 independent of t_1 . Therefore, its solutions cannot grow faster than an exponential on that interval. Consequently, x_5 remains bounded for all $t \in [0, t_1]$, and $\|\tilde{x}_e\|$ cannot escape to infinity on $[0, t_1]$. Since m_e, ρ_e, α_1 are independent of t_1 , we can repeat this argument for a new initial condition at t_1 and define the solution continuation in a new time interval $[t_1, 2t_1]$. This allows us to extend the time interval of the existence of solutions to the whole real axis, and conclude the proof. $\square$

5.2 Proof of Proposition 4.2

The first and second steps of this proof follow exactly the same arguments as above. In the third step we have the additional difficulty that now T_d is not a bounded external reference but is given as (4.5). To ensure (3.2) has a bounded right-hand side we included the projection operator in (4.6) and the assumptions of bounded $x_{5d}, \dot{x}_{5d}, \phi$. From here we conclude that x_5 cannot grow faster than an exponential on $[0, t_1]$, and proceeding as above we can extend the solutions to the whole real axis to conclude that $\tilde{x}_e$ is bounded and $\tilde{x}_e \rightarrow 0$ as $t \rightarrow \infty$.

We now proceed to prove asymptotic speed tracking. To this end, we replace (4.5) in (3.2) to get

$$J\dot{\tilde{x}}_5 + B\tilde{x}_5 = T - T_d + \tilde{p}^T \phi, \quad (5.6)$$

where we have defined the errors $\tilde{x}_5 \triangleq x_5 - x_{5d}$ and $\tilde{p} \triangleq \hat{p} - p$. Let us evaluate the derivative of the quadratic function

$$V \triangleq \frac{1}{2} J \tilde{x}_5^2 + \frac{1}{2g} \|\tilde{p}\|^2$$

along the trajectories of (5.6) and (4.6) to get

$$\dot{V} = -B\tilde{x}_5^2 + \tilde{x}_5(T - T_d) \quad (5.7)$$

$T - T_d$ bounded ensures via (5.7) that $\tilde{x}_5$ is bounded. From here, using (5.6), we conclude that also $\dot{\tilde{x}}_5$ is bounded, and consequently $\tilde{x}_5$ is uniformly continuous. Now, integrating (5.7) and using the fact that $T \rightarrow T_d$ as $t \rightarrow \infty$, we conclude that $\tilde{x}_5$ is square integrable. The proof is completed recalling that a uniformly continuous square integrable function converges to zero. $\square$

6. Concluding remarks

We have illustrated in this paper how by using energy shaping (passivity-based) design techniques we can obviate the need of state observers for the control of induction motors. This has been a long-standing problem, attracting the attention of many researchers, in both the practical and in the theoretical camps. We believe that this new application of the technique shows its potential and versatility, as well as underscores the importance of incorporating physical understanding into the controller design.

Our main motivation in carrying out this research was to address the combined parameter adaptation and state observation problem, which to the best of our knowledge is essentially open. As is well known, the

difficulty in that case stems from the fact that joint observation of states and parameters leads to bilinear estimation problems. It is our belief that 'observerless' schemes such as the one presented here provide a first step towards its solution, since the bilinearity is removed. Even though in this work we managed to solve a (restricted) load torque adaptation task, we were not able to address the parameter estimation problem. Some simple calculations show that, if we replace the parameter R_r by an estimate, we still get a linear in the parameters error equation. However, the measurements needed to design the update law include the rotor flux. Current research is under way to remove this difficulty.

As a concluding remark we would like to stress that the proposed controller can be obtained from a 'coordinate rotation' of the scheme in [11] plus a new damping injection mechanism to remove the need to measure (or observe) rotor flux. Consequently, as explained in [10], it is consistent with the field orientation idea extensively used in applications. Also, as seen from (5.3) it leads to steady-state balanced operation.

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